

System Architecture Overview

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> Version: 2026.0, Platform Engineering

> This document describes the production deployment

g and disaster recovery architecture for all customers.

1. Deployment Topology

1.1 Production is deployed on the public cloud across a single region.

1.2 Every service runs at least 3 replicas, spread across multiple availability zones.

1.3 The API gateway is the single entry point for all external requests over TLS.

1.4 Internal services communicate over gRPC; asynchronous data flows use Kafka.

1.5 The data platform uses ClickHouse for analytics and Elasticsearch for full-text search.

1.6 MySQL is used for transactional data that requires strong consistency.

2. Capacity Planning

2.1 The system is designed for a peak load of 50,000 requests per second.

2.2 A single service instance must not exceed 80% CPU utilisation under peak load.

2.3 Auto-scaling is triggered when CPU utilisation exceeds 80% for three consecutive minutes.

2.4 Database connection pools are sized at 100 connections per instance.

2.5 Cache clusters use Redis with 6 nodes in a 3-master configuration.

3. Disaster Recovery

3.1 Recovery Point Objective (RPO) is 15 minutes; Recovery Time Objective (RTO) is 4 hours.

3.2 Full database backups run daily; incremental backups run hourly.

3.3 Backup files are stored in object storage in a separate region.

3.4 Disaster recovery drills are conducted every quarter, involving all teams and led by the SRE team.

4. Middleware Selection

4.1 Kafka is used for event streaming; consumer groups are configured with idempotent semantics where supported.

4.2 Redis is used for hot caches with a maximum key size of 1MB.

4.3 Elasticsearch indices follow a daily rolling index lifecycle policy.

4.4 ClickHouse is used only for append-only analytics.